Bank Consolidation and Local Competition: Evidence from the Truist Merger

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Abstract

This paper studies how bank mergers affect deposit pricing and bank stability. We use the 2019 merger between BB&T and SunTrust, which formed Truist Financial Corporation, to identify the local competitive effects of consolidation in U.S. retail banking. Before the merger, the two banks overlapped in some counties but not others, generating variation in exposure to the merger across local markets and separating the competition effect from the scale effect of becoming a larger institution. Difference-in-differences event-study estimates show that deposit spreads rise in overlap counties relative to counties served by only one predecessor bank after the merger. The effects are larger in markets with weaker outside competition, but they are modest on average. We also estimate a structural deposit-demand model and use it to organize counterfactual exercises under alternative market structures. The counterfactual implies that the merger lowered deposit rates in overlapping counties at consummation, with the implied effect attenuating over the post-merger years, consistent with rate-sensitive depositor demand, a more centralized pricing, and competitive responses by other institutions. A complementary bank-level analysis of Z-scores suggests modest short-run effects on bank stability.

1 Introduction

Bank mergers create a tradeoff for regulators. Consolidation can reduce local competition, allowing banks to exercise greater market power over deposit rates. Larger banks, however, may also be more stable through economies of scale and diversification across geographies and loan portfolios. Understanding this competition–stability tradeoff is central to antitrust enforcement and prudential regulation, but empirical evidence on both sides of the tradeoff remains limited.

This paper studies this tradeoff in U.S. retail banking through the 2019 merger between BB&T and SunTrust, which formed Truist Financial Corporation. The merger created the sixth-largest bank in the United States and was the largest retail bank merger since the global financial crisis. The setting is useful for identification because BB&T and SunTrust had partial geographic overlap before the merger. Some local markets contained branches of both banks, while others contained only one. This structure creates variation that allows us to isolate changes in local competition from changes in bank scale.

We focus on deposit markets, which are central to both sides of the tradeoff. Deposits are the primary source of bank funding, accounting for the majority of liabilities for U.S. commercial banks. When competition is limited, banks can pay lower deposit rates and widen spreads relative to the policy rate. At the same time, insured deposits are a relatively stable funding source. Changes in deposit pricing and funding structure are therefore margins through which mergers affect both consumer welfare and financial stability.

We study how deposit pricing changes after the Truist merger across markets with different pre-merger exposure, using the Money Market $10,000 product on a bank-county-year market panel for the main difference-in-differences specification. Counties where both BB&T and SunTrust operated before the merger experienced an increase in local concentration; counties with only one bank did not. A difference-in-differences comparison of these groups measures how deposit spreads respond to increased market power while holding constant the effect of becoming part of a larger bank.

The reduced-form results show that rivals in overlapping counties widened their deposit spreads by approximately two basis points on the headline annual specification (Both vs. One contrast, money-market $10k product, 2016–2022, county and year fixed effects, 1.4 bp without controls and 2.1 bp with the full control set, significant at the 5% level in the latter), and by roughly 7–9 basis points at peak in the quarterly event study. The effect is small but persistent through the end of our sample period and consistent with parallel pre-trends in the leads. A branch-level robustness using only RateWatch data, which restricts the sample by outside competition, finds substantially larger effects in counties without a top-4 U.S. bank: in the Both-vs-One contrast on the branch-level RateWatch panel, deposit spreads in those markets widen by roughly 15 basis points after the

merger, consistent with reduced outside competitive pressure amplifying the local market-power effect.

The structural deposit-demand model on the headline money-market \$10,000 product recovers a positive and statistically significant own-rate coefficient ($\hat{\alpha} = 5.93$, $t = 3.41$) in the single-IV specification using log rival county footprint as the excluded instrument. The demand estimates feed into a no-merger counterfactual in which Truist's branches are reassigned to their pre-merger BB&T or SunTrust legacy entities and the equilibrium is re-solved. The implied rate effect on the merged firm at the 2019 baseline is a markdown of approximately 25 basis points, anti-competitive in sign and an order of magnitude larger than the reduced-form annual estimate. The implied change in depositor consumer surplus is large and negative (approximately \$556 million per year in overlapping counties), and the implied gain in Truist's producer surplus is approximately \$306 million per year. Forward simulations at each post-merger year show this anti-competitive rate effect attenuates from 2020 onward and reverses sign by 2021, suggesting the local market-power channel that drives the 2019 result fades as competitors adjust and the macro rate environment changes. The gap between the structural and reduced-form rate effects is consistent with the difference-in-differences netting out general industry trends that the structural model captures directly. On the stability side, a bank-level event study in log Z-score around the merger consummation shows a temporary drop concurrent with the COVID-19 pandemic followed by a recovery to the pre-merger trend by mid-2021. The pattern is consistent with operational integration friction at the moment of consummation followed by diversification benefits, although the overlap with COVID-19 makes a clean causal interpretation difficult.

We make three contributions. First, we provide clean reduced-form evidence on the deposit-pricing effect of a major regional bank merger using a within-merger difference-in-differences design that compares overlapping and non-overlapping counties served by the same merging institution. The partial-overlap design separates the local competition effect from the bank-wide scale effect. Second, we implement a no-merger counterfactual that re-imposes separate BB&T and SunTrust ownership using a 2019 Summary of Deposits branch-level legacy attribution. Most existing structural merger retrospectives evaluate the counterfactual at the pre-merger baseline year; we evaluate the counterfactual at each post-merger year using observed Truist prices and shares to recover marginal costs, producing a cleaner test of the static-logit framework's external validity over time. Third, we bring together pricing and solvency outcomes within the same merger event. While the literature has studied the competition-stability tradeoff theoretically and using cross-sectional variation, the partial-overlap design here lets us read pricing and solvency responses for the same set of banks in the same time window. The remainder of the paper is organized as follows. Sec-

tion 1.1 reviews the related literature. Section 2 describes the data, institutional background, and descriptive evidence. Section 3 presents the difference-in-differences design and reduced-form results. Section 4 develops the structural model. Section 5 reports the demand estimates and the no-merger counterfactual. Section 6 concludes.

1.1 Literature Review

This paper relates to three strands of literature: the competition-stability tradeoff, deposit market competition and pricing, and bank mergers and geographic pricing. It also draws on structural methods from industrial organization.

The competition-stability tradeoff has a long tradition in banking theory. Keeley (1990) argues that more competition erodes banks' charter value, reducing incentives for prudent behavior and increasing risk-taking. **?** push back, showing that lower competition can also raise risk through the loan-portfolio channel: when banks earn higher rents on performing loans, they have weaker incentives to monitor borrower risk. **?** reconcile the two views with a U-shaped relationship between competition and bank failure, in which both very competitive and very concentrated markets can produce high failure risk. Egan et al. (2017) link deposit competition to bank fragility through a structural model in which funding conditions affect risk choices, finding that deposit markets are a channel between competition and stability. Wang et al. (2022) find that bank market power dampens the transmission of monetary tightening. The 2023 banking turmoil, in which banks with large uninsured deposit shares experienced runs after interest rate increases (Jiang et al., 2023), provides more recent evidence that deposit market structure affects financial fragility. Our paper contributes to this strand by combining evidence on deposit pricing and Z-score solvency within the same setting.

A large literature studies how competition affects deposit pricing. Focarelli and Panetta (2003) show that bank mergers in Italy lead to higher spreads, particularly in markets where concentration increases. Craig and Dinger (2009) document how deposit rates change around U.S. bank mergers using product-level data. Bord (2018) studies how consolidation affects consumer financial welfare using county-level variation in merger exposure. **?** use the partial geographic overlap of a Canadian mortgage-market merger to identify local price effects, the same overlap-based identification strategy we apply here to U.S. deposit markets; **?** apply the same Canadian-merger design to identify the effect of bank consolidation on consumer bankruptcy through information loss. A common finding is that retail banking competition is local. Depositors place significant weight on branch proximity, and local bank presence shapes rates even in an era of large, multi-market banks (Nguyen, 2019). Structural work reaches a similar conclusion. Dick (2008), Ishii (2008), and Ho

and Ishii (2011) show that branch networks and bank size shape retail banking competition. Drechsler et al. (2017) provide a macro-level model in which banks' deposit market power shapes the transmission of monetary policy. Koont (2023) documents that digital banking has begun to erode the local nature of deposit competition. Our paper contributes by using the partial geographic overlap of two large regional banks to separate the competition effect of the merger from the scale effect.

A related literature studies how large banks set deposit rates across markets. Granja and Paixão (2024) show that large banks often apply uniform or zone-based rates across counties, and that mergers increase spreads most in markets where concentration rises most. This is relevant for the Truist setting, where both BB&T and SunTrust used standardized pricing. Begenau and Landvoigt (2022) note that uniform pricing complicates identification of the deposit channel, since banks may not respond to local conditions even when competitive pressure changes. Our difference-in-differences design addresses this by comparing changes within the same institution across markets with different overlap. In non-banking retail, Adams et al. (2019) find that finer geographic pricing raises profits; Labrador-Badia (2024) shows this result does not extend to banking, where competitive pressure limits the gains from price discrimination.

2 Data, Background, and Descriptives

This section describes the data, the institutional features of U.S. retail banking, the BB&T–SunTrust merger that motivates our empirical strategy, and descriptive statistics on the regression sample.

2.1 Data

Deposit rate data. Our primary source of deposit rate data is RateWatch, a commercial dataset that collects advertised deposit rates from individual bank branches across the United States. RateWatch provides detailed information on interest rates for a variety of common deposit products, along with branch identifiers and geographic location. RateWatch captures around 80% of U.S. bank branches.

We focus on rates for certificates of deposit (CDs) and money market accounts, which are widely available across banks and represent a substantial share of retail deposits. To construct a consistent measure of pricing, we compute deposit spreads as the difference between the federal funds rate and the offered deposit rate. This measure captures the extent to which banks pass through changes in policy rates to depositors and serves as a proxy for pricing power.

The data are at the branch-week level, which allows us to construct a panel of deposit rates across locations and time. For the main analysis, we aggregate the data to the county-month level, aligning with common definitions of local banking markets. The full data window is January 2012 through February 2025; specific analyses use shorter sub-windows that are noted in each section (the headline DiD uses 2016–2022 and the pre-merger demand sample is 2013–2019).

Branch-level deposits and market structure. We measure deposit quantities and market shares using the Summary of Deposits (SOD) dataset from the Federal Deposit Insurance Corporation (FDIC). The SOD provides annual data on total deposits held at each bank branch, along with detailed geographic information for the full universe of FDIC member banks.

We use the SOD to construct measures of local market structure, including the presence of BB&T and SunTrust prior to the merger and the degree of overlap between the two banks. The SOD also allows us to compute measures of market concentration, such as the Herfindahl-Hirschman Index (HHI), and to characterize the competitive environment in each local market.

Bank-level characteristics. We obtain bank-level financial information through bank call reports from the Federal Financial Institutions Examination Council (FFIEC). These data provide detailed information on bank balance sheets at the bank firm level, including total assets, deposits, capital ratios, and income measures. Using these data, we construct measures of bank size and financial health. In particular, we compute Z-scores for our measure of bank stability.

Macroeconomic data. We obtain the federal funds rate and other macroeconomic variables from the Federal Reserve Economic Data (FRED) database. The federal funds rate serves as the benchmark policy rate and is used to construct deposit spreads.

2.2 Institutional Background

We focus on three institutional features of U.S. retail banking: the role of deposits in bank funding, branch networks and local markets, and pricing practices across geographic markets.

Deposits and bank funding. Deposits are the primary source of funding for U.S. commercial banks, accounting for over 70% of bank liabilities. Compared to alternative funding sources such as wholesale borrowing, deposits are relatively low-cost, particularly insured deposits held by households. Because deposits are central to both bank profitability and financial stability, competition for deposits is a dimension of banking behavior.

Banks generate profits in part by earning a spread between the returns on their assets (such as loans and securities) and the interest rates they pay on deposits. When banks have market power in deposit markets, they can offer lower deposit rates and widen this spread. As a result, deposit pricing provides a direct measure of competitive conditions in retail banking.

Local markets and branch networks. Despite the presence of large, multi-market banks, retail banking remains local. Even in the era of digital banking, depositors place significant weight on geographic proximity when choosing a bank, with access to branches and ATMs a determinant of bank choice alongside rates and services (Nguyen, 2019; Dick, 2008). As a result, banks compete for deposits within geographically defined markets, typically at the level of counties or metropolitan areas. Banks operate networks of branches that determine their presence in local markets. Regulatory authorities, including the Department of Justice and the Federal Reserve, evaluate bank mergers using local measures of market concentration based on branch-level deposit shares.

By asset size, the banking industry is dominated by the four largest banks: JPMorgan Chase, Bank of America, Wells Fargo, and Citigroup. The U.S. has thousands of banks of varying sizes; appendix Table A.1 provides a full snapshot of the universe of U.S. commercial banks from the FDIC Summary of Deposits, broken down by AUM tier, state footprint, and county footprint. Before the Truist merger, SunTrust and BB&T were in the \$100–\$250 billion tier; by merging, they jumped into the \$250 billion to \$1 trillion tier. Table 1 shows the corresponding tier breakdown for the subset of banks observed in our regression sample, with deposit interest rates and county-level statistics relevant to the empirical analysis below.

Deposit pricing and geographic pricing policies. Deposit pricing in modern banking is not set independently at each branch. Instead, banks typically establish networks of branches and adopt centralized pricing policies within those networks, setting deposit rates uniformly across large geographic areas or within defined pricing zones. These zones may include multiple counties or entire states, and rates are often determined by a subset of "rate-setting" branches that set prices for "rate-taking" branches.

This pricing structure reflects both operational considerations and competitive strategy. Setting rates centrally reduces administrative complexity and ensures consistency across a bank's network. At the same time, uniform or zone-based pricing constrains banks' ability to tailor rates to local competitive conditions. As a result, observed deposit rates reflect a combination of local market structure and bank-wide pricing decisions.

This feature matters for interpreting the effects of mergers. When a merger changes local market concentration, the resulting pricing response may occur through adjustments in bank-wide pricing

policies rather than purely local changes.

Implications for empirical analysis. These institutional features motivate our empirical approach. First, because competition operates at the local level, we define markets geographically and exploit variation in local overlap between merging banks. Second, because pricing is often centralized, we interpret changes in deposit rates as reflecting both local competitive conditions and bank-level pricing strategies.

2.3 The Truist Merger

The 2019 merger between BB&T and SunTrust provides a natural setting to study the effects of consolidation in retail banking. Announced in February 2019 and completed in December 2019, the merger combined two large regional banks to form Truist Financial Corporation, the sixth-largest bank holding company in the United States by assets.

BB&T and SunTrust had substantial but incomplete geographic overlap prior to the merger. Both banks operated extensive branch networks across the southeastern United States, but their presence varied across local markets. Some counties contained branches of both banks, while others contained only one. This variation is central to our empirical strategy.

Overlap and changes in local competition. In markets where both BB&T and SunTrust operated prior to the merger, the transaction reduced the number of competitors by combining two local banks into one. These markets experienced an increase in local concentration. In contrast, in markets where only one of the two banks was present, the merger did not directly change local concentration, even though the bank became part of a larger institution.

Truist primarily operates in the Southeast, and it is headquartered in Charlotte, NC. Figure 1 shows the geographic distribution of BB&T and SunTrust prior to the merger. Out of 507 affected counties, 184 contained branches of both banks, while the remainder had only one. The distinction between overlapping and non-overlapping counties allows us to separate two effects of the merger. The first is a *competition effect*, arising from increased concentration in overlapping markets. The second is a *scale effect*, arising from changes in bank size, funding, and operations that affect all markets in which the merged bank operates. By comparing overlapping and non-overlapping markets, we isolate the impact of changes in competition while holding constant broader changes in bank scale.

After the merger, Truist closed around a third of their approximately 3,000 branches. Figure B.5 shows the evolution of branch networks over time, indicating substantial post-merger consolidation.

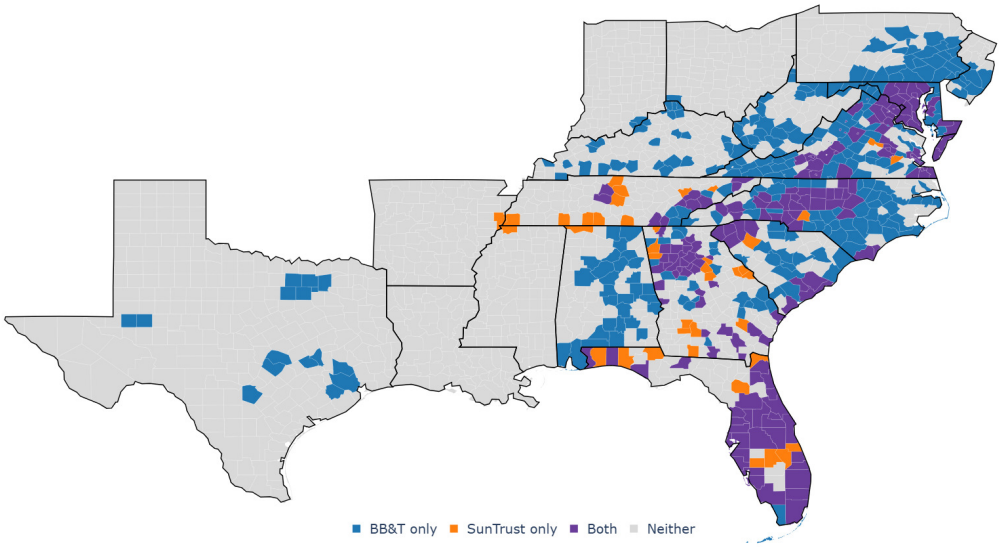


Figure 1: Pre-merger presence of BB&T and SunTrust by county

Notes: As of June 30, 2019. Counties in purple had branches of both BB&T and SunTrust before the merger. Counties in blue and orange had branches of either BB&T or SunTrust, respectively, but not both. Grey counties had neither bank. Data source: Summary of Deposits.

The decrease in the number of Truist branches happens for two main reasons. First, there were a number of voluntary branch consolidations post-merger when branches of BB&T and SunTrust were nearby. These branches combined to operate a larger branch while still maintaining the same total deposits and similar geographic footprint. We argue this does not significantly impact our empirical strategy because Truist is still gaining deposit market share, even if it operates out of fewer total branches in the county. Of the 1,054 total Truist branch closures after the merger, 1,026 were voluntary closures. Second, a small handful of branch closures were required by the Federal Reserve as terms to approve the merger. These were cases where both banks had a relatively large presence within the same small counties. For these counties, Truist divested the existing SunTrust branches to competitors who had little or no presence in that county. This is the case for only 28 branches, which does not meaningfully affect our analysis.

Even with these closures, the merger results in a large increase in deposit market concentration in many counties. Figure B.6 shows the distribution of changes in deposit HHI across the 184 overlapping counties, with most markets experiencing sizable increases in concentration following the merger. Under the DOJ's current Bank Merger Guidelines, an increase in HHI of 100 or more triggers heightened antitrust scrutiny. More than half of the overlapping counties had their deposit HHI increase by more than 100 after the merger. These figures show that the merger reduced local competition in overlapping markets and reshaped the branch network in ways consistent with

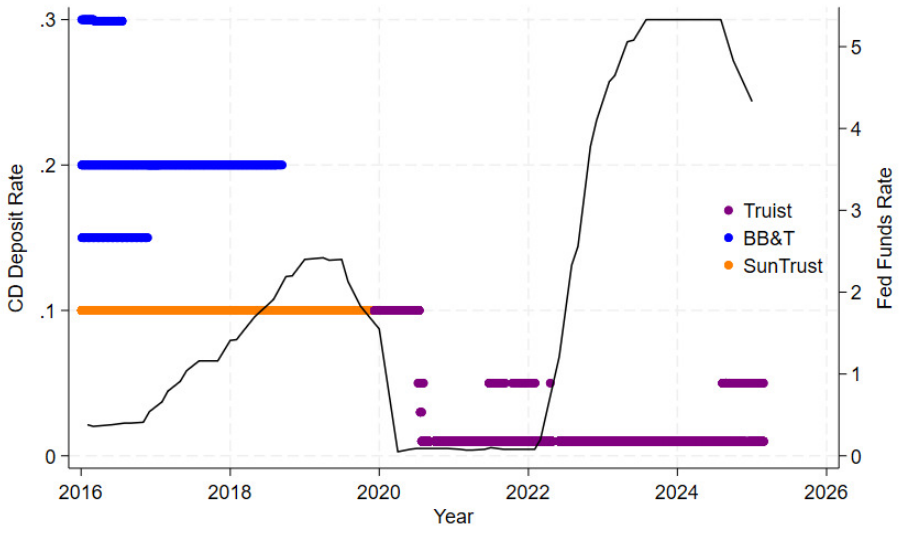


Figure 2: CD deposit rates and the federal funds rate

Notes: Average deposit rates on 12-month certificates of deposit with a $10k minimum balance, by bank branch and month for branches of SunTrust, BB&T, and Truist. The black line represents the federal fund rate. The deposit spread (federal funds rate minus deposit rate) is the main outcome variable. The vertical line marks December 2019 (merger completion). Data sources: RateWatch, Federal Reserve Economic Data.

increased market power.

Impact on deposit pricing. Standard models of deposit competition predict that an increase in local concentration reduces competitive pressure and allows banks to lower deposit rates, widening spreads relative to the policy rate. In the Truist setting, this effect may operate locally, through reduced competition in overlapping counties, and centrally, through adjustments in bank-wide pricing policies. Figure 2 shows CD deposit rates for SunTrust, BB&T, and Truist alongside the federal funds rate. Each point is the deposit rate offered by a bank branch. Before the merger, BB&T and SunTrust offered rates that were higher and more dispersed across branches than Truist's rates after the merger. This change in pricing may reflect a more uniform pricing strategy, an increase in market power, or both. Deposit rates remained low even as the Federal Reserve raised rates during the post-merger period.

Relevance for financial stability. The merger also substantially increased the size of the combined institution. A larger bank can pool risk across geographic markets and loan portfolios and exploit economies of scale, channels through which scale may improve solvency. Operational integration costs in the first quarters after consummation can offset these benefits in the short run. We examine the net effect on Truist directly in the bank-level Z-score event study in Section 3.5.

At the same time, changes in deposit pricing and funding structure may affect the composition

of bank liabilities, with implications for risk. For example, lower deposit rates may reduce funding costs but could also lead to shifts in depositor behavior or increased reliance on alternative funding sources.

Table 1: Regression-sample bank and county characteristics, June 2019 (pre-merger)

Panel A. Bank size by assets under management (AUM)

	Number of Firms	Number of Branches	Deposits Held ($B)	Deposit Share	MM$10k Rate (bp)
$1+ T	4	15,260	$3,213	47.3%	14.6
$250 B – $1 T	5	6,348	$775	11.4%	25.7
$100 B – $250 B	14	9,651	$1,130	16.6%	18.5
$10 B – $100 B	84	11,638	$996	14.7%	20.3
$1 B – $10 B	415	10,240	$459	6.7%	22.8
≤ $1 B	1,332	8,081	$222	3.3%	25.8
Total	**1,854**	**61,218**	**$6,795**	**100.0%**	

Panel B. Bank size by state footprint

	Number of Firms	Number of Branches	Deposits Held ($B)	Deposit Share	MM$10k Rate (bp)
25+ states	3	12,353	$2,060	30.3%	3.5
10-24 states	9	14,755	$2,162	31.8%	19.6
2-9 states	257	19,167	$1,893	27.9%	31.3
1 state	1,585	14,943	$679	10.0%	24.7

Panel C. County footprint of banks operating in one state

	Number of Firms	Number of Branches	Deposits Held ($B)	Deposit Share	MM$10k Rate (bp)
10+ counties	9	592	$39	5.7%	44.5
2-9 counties	975	10,531	$567	83.4%	23.5
1 county	601	3,820	$74	10.9%	22.2

Panel D. County-level: characteristics by number of banks operating in the county

	Number of Counties	Mean # Banks/Cnty	Deposits Held ($B)	Deposit Share	MM$10k Rate (bp)
10+ banks	152	11.0	$1,554	22.9%	19.3
5-9 banks	791	6.7	$4,604	67.8%	19.7
3-4 banks	382	3.5	$612	9.0%	10.3
2 banks	147	2.0	$17	0.3%	21.6
1 bank (monopoly)	113	1.0	$8	0.1%	24.0
Total	**1,585**		**$6,795**	**100.0%**	

Notes: Pre-merger snapshot at June 30, 2019, restricted to bank-county-year cells observed in our Money Market $10,000 RateWatch panel (the structural-demand and headline DiD sample). Panels A–C aggregate to the bank level (1,854 unique banks); Panel D aggregates to the county level (1,585 unique counties). The smaller bank universe (vs. 4,649 in the full SoD snapshot in appendix Table A.1) reflects RateWatch coverage of the MM$10k product. "MM$10k Rate" is the deposit-weighted mean Money Market $10,000 12-month APY in basis points: in Panels A–C it is first computed within each bank as the deposit-weighted mean across the bank's branches (using branch deposits as weights), then aggregated within each tier as a deposit-weighted mean across banks. In Panel D it is first computed within each county as the deposit-weighted mean across banks operating in the county, then aggregated within each tier as a deposit-weighted mean across counties. Deposit share in Panels A, B, D is computed as a fraction of the regression-sample total; in Panel C it is computed relative to single-state banks only. Panel C is restricted to banks operating in a single state. Panel D bins counties by the number of banks operating at least one branch in the county at June 2019. BB&T and SunTrust appear in the $100B–$250B tier of Panel A. Data sources: Summary of Deposits (FDIC), RateWatch, FFIEC Call Reports.

3 The Effect of the Merger on Deposit Spreads

This section presents the difference-in-differences design that compares counties where both BB&T and SunTrust operated before the merger to counties served by only one of the two, the headline reduced-form estimates of the merger's effect on deposit spreads, and a complementary bank-level event study of financial stability.

3.1 Empirical specification

We estimate how the Truist merger affected deposit spreads using a difference-in-differences design. We define the treatment variable based on the way the merger impacted each county. In our primary analysis, the treatment variable is whether a county had branches of *both* BB&T and SunTrust before the merger, and we compare these against counties with only *one* of the two. The treated counties experienced an increase in local concentration when the two banks merged; counties with only one of the pre-merger banks did not. The outcome is the deposit spread, defined as the federal funds rate minus the deposit rate offered at a given branch j in market m and month t:

$$\text{Spread}_{jmt} \equiv (\text{Fed Funds Rate})_t - (\text{Deposit Rate})_{jmt}. \tag{1}$$

A higher spread means the bank pays a lower rate relative to the policy rate, indicating greater pricing power.

The current implementation uses the Money Market \$10,000 annual bank–county market panel and excludes the merging banks when estimating rival-bank pricing responses. Treatment is assigned at the county level. For bank j in county m and year t, we estimate

$$\text{Spread}_{jmt} = \alpha + \beta\,(\text{Post}_t \times \text{Treat}_m) + X'_{jmt}\Gamma + \mu_j + \mu_m + \tau_t + \varepsilon_{jmt}, \tag{2}$$

where $\text{Post}_t = 1$ in the post-merger period, $\text{Treat}_m = 1$ if market m had branches of both BB&T and SunTrust before the merger, μ_j are bank fixed effects, μ_m are county fixed effects, and τ_t are year fixed effects. The vector X_{jmt} includes bank-level controls (log total branches, log bank-county footprint, branch share within the county, and a large-bank indicator) and market-structure controls (county Herfindahl index, branch share within the county, and RateWatch coverage of the county's deposit base). Standard errors are clustered at the bank-county level. The coefficient β measures how rival-bank spreads change in treated markets relative to untreated markets after the merger, holding fixed bank, county, and year differences.

Table 2: Treatment and control groups for reduced-form analysis

Specification	Treated counties	Control counties	Focus of comparison
Both vs. One *(preferred)*	Had both BB&T and SunTrust	Had one bank only	Concentration
One vs. None	Had one bank only	Had neither bank	Scale
Any vs. None	Had at least one bank	Had neither bank	Geographic footprint
Both vs. None	Had both BB&T and SunTrust	Had neither bank	Broad treated vs. untreated

Notes: Treatment status is based on 2019 Summary of Deposits data. "Both" counties had branches of both BB&T and SunTrust before the merger. "One" counties had either BB&T or SunTrust, which became part of Truist, but their local number of competitors did not change. "None" counties had no pre-merger presence of either bank.

3.2 Treatment and control groups

Table 2 describes the four comparisons we use in our estimation. The preferred comparison is "Both vs. One" described above: counties where both banks were present (treated) against counties where only one was present (control). Both groups become part of Truist after the merger, so the comparison holds the bank-scale effect constant and isolates the competition effect.

In our other specifications, we make use of the "None" counties, which had neither BB&T nor SunTrust before the merger and were thus not directly affected by the change in concentration or the scale effect. Table 2 lays out the other three comparisons we make, which decompose the forces behind the merger: the scale effect ("One vs. None"), the geographic footprint effect ("Any vs. None"), and a broad treatment-vs-untreated comparison ("Both vs. None").

3.3 Identification

The identifying assumption is parallel trends: deposit spreads in overlapping counties would have followed the same trend as spreads in non-overlapping counties absent the merger. Because both groups belong to the same institution post-merger, and face the same bank-wide pricing policies, this assumption is more credible than comparisons across different banks. We examine pre-trends using an event-study version of equation (2) at the quarterly level using the Money Market $10,000 market panel:

$$Spread_{jmt} = \alpha + \sum_{k \neq -1} \beta_k \left(\mathbf{1}\{t - t_0 = k\} \times \text{Treat}_m \right) + \mu_m + \tau_t + \varepsilon_{jmt}. \tag{3}$$

Here, t_0 denotes the merger quarter (2019Q4), and the omitted period is $k = -1$, so coefficients β_k are treatment effects interpreted relative to the quarter immediately preceding the merger.

3.4 Reduced-form results

Balance tests. Table B.4 reports summary statistics and pre-merger balance tests across treatment and control groups for all four specifications. Average deposit spreads in all four groups of counties are just over 0.9 percentage points. The "None" counties (where Truist does not operate before or after the merger) have notably higher deposit spreads and fewer firms competing against each other. We further evaluate the identifying assumption using the event study specification in equation 3. The pre-merger coefficients are small, statistically insignificant, and show no clear trend across all four contrasts (Figure B.1), providing evidence in favor of parallel pre-trends.

Main difference-in-differences estimates. Table 3 reports the main annual market-panel difference-in-differences specification (equation 2) for the Both vs. One contrast on the Money Market $10k product. The two columns add ingredients sequentially: county and year fixed effects, then bank and market controls. Standard errors are clustered at the bank-county level. The deposit spread used as the dependent variable is the federal funds rate minus the offered deposit rate; pre-merger average spreads across the four contrast groups are just over 90 basis points (Table B.4).

The fixed-effects-only specification (column 1) yields a treatment effect of 1.4 basis points (significant at the ten-percent level), and adding the full control set (column 2) raises this to 2.1 basis points (significant at the five-percent level). We read the headline annual effect as approximately two basis points of additional spread in overlapping counties relative to one-Truist counties. Appendix Table B.1 reports the same regression with the more restrictive bank-county fixed effects, where the within-bank-county variation is too narrow to identify a precise effect; we treat the county-and-year fixed-effect specification as the headline because it preserves cross-bank variation within counties.

Quarterly event study. The annual specification averages each year's deposit rate over twelve months centered on June, which mechanically dampens any short-run response to the merger because the year-2020 observation mixes pre- and post-merger quarters. We therefore turn to the quarterly market-panel event study in equation 3, which estimates a separate coefficient for each post-merger quarter and provides a cleaner reading of the dynamic effect. Appendix Figure B.1 (Money Market $10,000) and Appendix Figure B.2 (Money Market $2,500) plot the dynamic event-study coefficients for the Both vs. One contrast. The pre-merger coefficients are small and statistically insignificant, consistent with parallel pre-trends. After the merger the point estimates rise to roughly seven to nine basis points in the first six post-merger quarters in the $10,000 panel and remain elevated through quarter twelve, and to a smaller magnitude in the $2,500 panel consistent

Table 3: Both-vs-One Annual Market-Panel DiD, Money Market $10,000 (county and year fixed effects, no bank fixed effects)

	(1) FE only	(2) + controls
Treat × Post	0.014*	0.021**
	(0.008)	(0.008)
County FE	✓	✓
Year FE	✓	✓
Bank + Market Controls		✓
Observations	6,817	6,817

Notes: Annual market-panel difference-in-differences estimates for the Both vs. One contrast on the Money Market $10,000 product. Sample is bank–county–year cells for the period 2016–2022, restricted to rivals of BB&T and SunTrust (the merging banks themselves are excluded). Dependent variable: deposit spread, defined as the federal funds rate minus the deposit rate, in percentage points. Treat equals one in counties where both BB&T and SunTrust operated before the merger and zero in counties served by only one of the two; Post equals one for years 2021 onward, with 2020 dropped as a transition year because the trailing-twelve-month rate measure for that year averages pre- and post-December-2019 pricing. Both columns include county and year fixed effects. Column (1) reports the baseline specification with fixed effects only. Column (2) adds the bank-level controls (log of total branches, log of bank-county footprint, branch share within the county, large-bank indicator) and market controls (county Herfindahl index, branch share within the county, RateWatch coverage of the county's deposit base). Standard errors are clustered at the bank-county level and reported in parentheses. Two additional specifications, (3) bank-deposit weighting and (4) Truist exposure weighting, were estimated and produced larger point estimates with wider confidence intervals; they are not reported here because the weighted standard errors are inflated by a small number of large-deposit cells. The bank-county fixed-effect counterpart of this regression is reported in Appendix Table B.1 as a robustness exercise. Data sources: RateWatch, Summary of Deposits, Federal Reserve Economic Data, FFIEC Call Reports. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

with the higher rate sensitivity at the lower-balance tier. The effect is statistically significant at the ten-percent level in several post-merger quarters and persists at similar magnitudes throughout the post-period.

Magnitudes and interpretation. The quarterly estimates indicate that rivals in overlapping counties widened their spreads by approximately seven to nine basis points after the merger relative to rivals in one-Truist counties, while the four-year average estimate from the annual specification is approximately two basis points. The two are consistent: averaging quarterly effects over a window that includes 2020 (a year in which the trailing-twelve-month rate measure mechanically blends pre- and post-merger pricing) attenuates the annual estimate. Both magnitudes are smaller than the macro-level estimates of the deposit channel of monetary policy reported in Drechsler et al. (2017) but in line with the ten-basis-point range reported in retail-banking merger retrospectives such as Focarelli and Panetta (2003), Craig and Dinger (2009), and Bord (2018).

Two features of the design help interpret the magnitudes. First, both BB&T and SunTrust used standardized, zone-based deposit pricing before the merger. Bank-wide pricing rules attenuate the local response to a change in local concentration, since the same posted rate is shared across many

counties. Second, depositor responsiveness to small rate differences appears high in money market products, which is consistent with the elastic demand we recover in the structural model below. Both forces compress the price effect of any single market-structure change.

Auxiliary specifications. The Both vs. None and One vs. None contrasts compare overlapping or single-Truist counties to counties without any Truist branch. These contrasts are not the cleanest test of competition because the no-Truist counties differ in observable characteristics, but they help disentangle the local competition effect from a bank-wide "scale" effect (One vs. None) and from a geographic footprint effect (Any vs. None). We report them in Appendix Table B.5, which uses the raw branch-level RateWatch panel and a leaner control set (only month and county fixed effects). The Both vs. None contrast yields point estimates that are similar in sign but consistently smaller in magnitude and statistically indistinguishable from zero, confirming that the within-Truist contrast in Table 3 is the more informative comparison. We also report event-study estimates for the Money Market \$2.5k product in Appendix Figure B.2; the post-merger pattern is similar to the \$10k version with smaller magnitudes.

This motivates the structural approach developed in the next section. A model of deposit demand and supply allows us to (i) quantify the degree of market power implied by observed pricing behavior, (ii) separate the competition effect from the scale effect in a unified framework, and (iii) run a no-merger counterfactual that recovers what deposit rates would have been under separate BB&T and SunTrust ownership.

3.5 Stability effects

We measure solvency risk using the bank Z-score

$$Z_{jt} = \frac{\text{ROA}_{jt} + E_{jt}/A_{jt}}{\sigma(\text{ROA}_j)}, \tag{4}$$

with $\sigma(\text{ROA}_j)$ a four-quarter rolling standard deviation. We work in logs because the distribution is right-skewed, and estimate

$$\log Z_{jt} = \alpha + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - t_0 = k\} \cdot \text{Truist}_j + \mu_j + \tau_t + \varepsilon_{jt}, \tag{5}$$

on a panel of 128 regional banks with 2019 assets between \$10 billion and \$1 trillion. t_0 is 2019Q4 and Truist_j pools BB&T, SunTrust, and post-merger Truist. Standard errors are clustered at the bank level.

Appendix Figure B.3 shows the estimated dynamic path. At the merger quarter Truist's log

Z-score drops by approximately 1.9 log points relative to the control group; from quarter 3 onward the effect reverses to roughly 0.7–0.9 log points and fades to zero by mid-2021. We read this as integration friction at consummation followed by a diversification benefit. We treat the evidence as descriptive rather than causal: the COVID-19 shock in 2020Q1–Q2 mechanically lowered nearly all banks' Z-scores by raising aggregate ROA volatility, and the result is sensitive to the choice of pre-merger treatment identifier (BB&T-only, SunTrust-only, or pooled), which we report in Appendix Figure B.4.

The pre-merger balance tests for Z-score and its component variables are reported in Table B.6. On average, SunTrust and BB&T were smaller and had lower Z-scores than their peer banks before the merger. However, they were similarly profitable and well-capitalized, as shown by their return on assets and equity-to-assets ratio that closely align to the comparison group.

4 Model

This section develops a model of deposit demand that allows us to estimate demand elasticities in this market and conduct counterfactual analysis.

4.1 Deposit demand

We model depositors as choosing among banks operating in a local market. A market is a county, indexed by m, and time t is the year (or year-quarter in the quarterly panel). The demand model is estimated separately for each deposit product (the headline analysis uses the Money Market \$10,000 account); within a product, the unit of observation is a bank-county-year cell, with each bank j's rate aggregated (deposit-weighted) across its branches in the county.

Each bank j in market m at time t offers its deposit product at an interest rate r_{jmt}, with observable bank-county characteristics x_{jmt}. These characteristics include measures of branch presence and other bank attributes. In addition, depositors may choose an outside option, which captures alternatives such as holding cash, using non-bank financial institutions, or selecting banks not included in the sample.

Bank solvency risk. Before specifying utility, we use the same bank-level Z-score defined in equation (4), which captures distance to insolvency. A higher value indicates a bank further from insolvency.

Utility specification. The utility that depositor i in market m at time t derives from choosing bank j is

$$u_{ijmt} = \alpha r_{jmt} + \beta_b b_{jmt} + \beta_z \log Z_{jt} + x'_{jmt}\gamma + \xi_{jmt} + \varepsilon_{ijmt}, \tag{6}$$

where r_{jmt} is the deposit interest rate, b_{jmt} is a measure of branch presence, Z_{jt} is the bank Z-score defined in equation (4), and x_{jmt} is a vector of other observable bank characteristics. The term ξ_{jmt} captures unobserved product quality, and ε_{ijmt} represents idiosyncratic preferences assumed to be i.i.d. Type I extreme value.

The parameter α captures depositors' sensitivity to interest rates. β_b captures the value of branch convenience. β_z captures depositors' preference for banks with lower default risk: following Egan et al. (2017), depositors may prefer banks further from insolvency, particularly for uninsured balances.

Market shares. Under the assumption that ε_{ijmt} follows a Type I extreme value distribution, the model yields standard logit demand. The market share of bank j in market m at time t is given by

$$s_{jmt} = \frac{\exp(\delta_{jmt})}{1 + \sum_{k \in \mathcal{J}_{mt}} \exp(\delta_{kmt})}, \tag{7}$$

where

$$\delta_{jmt} = \alpha r_{jmt} + \beta_b b_{jmt} + \beta_z \log Z_{jt} + x'_{jmt}\gamma + \xi_{jmt} \tag{8}$$

is the mean utility of bank j, and the denominator includes all banks in the market as well as the outside option.

The outside option share is given by

$$s_{0mt} = \frac{1}{1 + \sum_{k \in \mathcal{J}_{mt}} \exp(\delta_{kmt})}. \tag{9}$$

Estimating equation. Following Berry (1994), we can invert the market share equation to obtain the estimating equation

$$\log(s_{jmt}) - \log(s_{0mt}) = \alpha r_{jmt} + \beta_b b_{jmt} + \beta_z \log Z_{jt} + x'_{jmt}\gamma + \mu_m + \tau_t + \xi_{jmt}. \tag{10}$$

In the baseline estimates, μ_m are county fixed effects and τ_t are year fixed effects. The branch variable b_{jmt} is measured as log branch count in the county, and $\log Z_{jt}$ is the natural log of the four-quarter rolling Z-score defined in equation (4).

In this framework, deposit rates play the role of prices: α describes the elasticity of demand

with respect to rates, while β_b captures non-price competition through branch convenience. The unobserved term ξ_{jmt} absorbs persistent differences in bank appeal (brand strength, service quality) that would otherwise bias the rate coefficient.

4.2 Deposit supply and bank pricing

We adopt a standard oligopoly framework in which banks compete in deposit rates and set them to maximize variable profits, taking the rates of other banks as given.

Bank j operates across a set of local markets $\mathcal{M}_j$. Under price discrimination, the bank chooses a separate deposit rate r_{jmt} in each market $m \in \mathcal{M}_j$ at time t to maximize total variable profit:

$$\pi_{jt} \;=\; \sum_{m \in \mathcal{M}_j} \big(l_{jt} - r_{jmt} - mc_{jmt}\big) \cdot s_{jmt}\big(r_{jmt},\, r_{-j,mt}\big) \cdot D_{mt}, \tag{11}$$

where l_{jt} is bank j's return on the assets that deposits fund, mc_{jmt} is the marginal cost of servicing a unit of deposits in market m (operational costs, FDIC insurance, reserve requirements), $s_{jmt}(\cdot)$ is the logit market share from the demand model, and D_{mt} is the size of the local deposit market. The bank earns the per-dollar margin $l_{jt} - r_{jmt} - mc_{jmt}$ on every dollar of deposits it attracts in market m. We assume that the marginal return l_{jt} does not respond to small changes in any single market's deposit rate, following Ho and Ishii (2011) and the assumptions used in Labrador-Badia (2024), and that competition is local so that rates in one market do not directly affect demand in another.

Under price discrimination the first-order condition for r_{jmt} decouples market-by-market:

$$\big(l_{jt} - r_{jmt} - mc_{jmt}\big) \cdot \frac{\partial s_{jmt}}{\partial r_{jmt}} \;=\; s_{jmt}, \tag{12}$$

which rearranges to the markdown condition

$$l_{jt} - r_{jmt} \;=\; mc_{jmt} \;+\; \frac{s_{jmt}}{\partial s_{jmt}/\partial r_{jmt}}. \tag{13}$$

This is the deposit-bank analogue of the standard logit markup. Because $\partial s_{jmt}/\partial r_{jmt} > 0$ (depositors prefer higher rates), the right-hand correction is a positive markdown: the bank pays less than its full opportunity cost $l_{jt} - mc_{jmt}$, with the size of the wedge inversely proportional to the local rate sensitivity of depositor demand. Banks with more captive depositors (smaller $\partial s_{jmt}/\partial r_{jmt}$) pay lower rates and earn larger spreads relative to the policy rate. Increases in local concentration following a merger reduce $|\partial s_{jmt}/\partial r_{jmt}|$, raise the markdown, and widen spreads.

In practice, both BB&T and SunTrust used standardized zone-based pricing rather than setting

a separate rate in each county. We treat the market-by-market specification as a benchmark and discuss zone-pricing constraints, which would attenuate the implied response to local concentration, in the discussion of the counterfactual in Section 5.2.

5 Structural Results and Counterfactual Analysis

This section reports the demand estimates from the model in Section 4 and uses them in a no-merger counterfactual that recovers what deposit rates would have been if BB&T and SunTrust had remained separate.

5.1 Demand estimation

Identification. Deposit rates are endogenous to local demand because a bank that anticipates a shock to county-level deposit demand can adjust its offered rate at the same time, attenuating the OLS estimate of α. We instrument the rate using the leave-one-out log of the number of counties served by rival banks in the same market, following the BLP-style rival-characteristic logic of Berry (1994). Rivals' geographic scope shifts a bank's effective competitive environment, both through how many alternative banks a depositor in the county faces and through the rivals' incentive to offer a uniform rate across their pricing zone, without entering the depositor's utility for the focal bank directly. The fixed effects absorb time-invariant county-level differences in deposit demand and aggregate rate movements over time, so identification comes from within-county, within-year variation in relative competitive conditions across banks.

The baseline demand specification is estimated on the pre-merger 2013–2019 sample for the Money Market $10,000 product. The deposit-rate variable is the deposit-weighted mean rate across each bank's branches in the county; the dependent variable is the logit mean utility $\delta_{jmt} = \log(s_{jmt}) - \log(s_{0mt})$. The headline column uses a single just-identified instrument (log rival county footprint, the leave-one-out log of the number of counties served by rival banks in the same market). For robustness we report two additional columns: a specification that augments the headline IV with operating-cost shifters and a nested-logit specification with within-nest share. The headline estimates are reported in Appendix C.1 (Table C.1).

The IV column produces a positive and statistically significant coefficient on the deposit rate, consistent with the standard logit demand model in deposits: depositors prefer higher rates. The branch-network variables enter with the expected signs and are precisely estimated. Branch presence enters in logs, so the coefficient on log branches in county is the elasticity of $\log(s_{jt}/s_{0t})$ with respect to branch count: a coefficient of approximately 0.73 implies that a 10% increase in a bank's

branches in the county raises the inside-versus-outside share ratio by about 7%. The estimated rate coefficient $\hat{\alpha}$ in the IV column implies an average own-rate elasticity of demand at the 2019 baseline of $\hat{\alpha}\,(1-\bar{s}_{jm})\,\bar{r}_{jmt}$, computed at observed shares and rates. We use the IV column as the headline specification for marginal-cost recovery and the counterfactual exercise in Section 5.2; the cost-shifter and nested-logit columns serve as robustness.

5.2 Counterfactual analysis

We use the estimated demand and supply model based on the Money Market $10,000 baseline to simulate counterfactual deposit rates under a no-merger ownership structure. The counterfactual is designed to answer two questions: (1) How much of the observed increase in deposit spreads in overlapping counties is attributable to the competition effect of the merger, as opposed to other changes in bank behavior? (2) What are the implied welfare effects for depositors?

In each counterfactual, we hold the estimated demand parameters and recovered marginal costs fixed at their pre-merger values and compute the new Nash equilibrium deposit rates by iterating the system of first-order conditions to convergence. We report both rate changes relative to the observed post-merger equilibrium and the implied change in depositor consumer surplus using the standard logit welfare formula.

5.3 No-merger benchmark

The first counterfactual asks: what would deposit rates have been in the absence of the merger? We simulate a counterfactual market structure in which BB&T and SunTrust continue to operate as independent banks in all markets, including overlapping counties, with their pre-merger demand parameters and marginal costs. We then solve for the Nash equilibrium rates and compare them to the observed post-merger rates for Truist. The exercise corresponds to the limiting case of a full divestiture remedy in which all of Truist's overlapping-county branches are reassigned to their pre-merger owners, and provides the policy-relevant benchmark for evaluating the merger's competitive effect. The Money Market $10,000 baseline produces the cleanest demand estimate, so we use it as the main simulation benchmark and omit the CD diagnostics from the main text.

Formally, at each post-merger year t we hold the demand parameters fixed at their pre-merger estimates, reassign Truist's branches to their BB&T or SunTrust legacy entities, and re-solve:

$$r_{jmt}^{\text{no-merger}} = l_{jt} - mc_{jmt} - \frac{s_{jmt}^{\text{no-merger}}}{\partial s_{jmt}^{\text{no-merger}} / \partial r_{jmt}}, \tag{14}$$

for all banks j simultaneously. The difference $r_{jmt}^{\text{observed}} - r_{jmt}^{\text{no-merger}}$ isolates the competition effect of

Table 4: No-merger counterfactual rate change by year, Money Market $10,000 (headline demand specification, column 1 of Table C.1)

	2019 (headline)	2020	2021	2022	2023
Δr_{merged} (bp)	**−25.5**	−8.7	+4.8	+5.3	+9.5
N obs	50,242	∼33,000	∼33,000	∼33,000	∼33,000
Converged	1	1	1	1	1

Notes: Static (column 1) and forward (columns 2–5) counterfactuals at one year each. The 2019 column re-solves the un-merger at the actual 2019 baseline panel and is the headline structural number used in the paper text and Table 5. The 2020–2023 columns use the 2019-Summary-of-Deposits legacy attribution to artificially split Truist into BB&T and SunTrust legacy entities and re-solve at each post-merger year, holding pre-merger demand parameters fixed at the 2013–2019 estimates ($\hat{\alpha}$ = 5.93 pp, SE = 1.74, t = 3.41, Kleibergen–Paap F = 22.9). The forward years should be read as exploratory: depositor preferences over rate vary with the level of the federal funds rate, which spiked in 2022–2023, so $\hat{\alpha}$ held fixed at the pre-merger value is a strong assumption out of sample. Negative Δr_{merged} means the merger lowered Truist's rate (anti-competitive); positive means the merger raised it. Data sources: RateWatch, Summary of Deposits, Call Reports.

the merger on deposit rates in each market.

Demand parameters (α, β) are estimated on the pre-merger 2013–2019 sample under the single-IV specification reported in Table C.1 and held fixed across the post-merger years; the counterfactual is reported at the 2019 baseline as the headline (Table 5) with each post-merger year $t \in \{2020, 2021, 2022, 2023\}$ available as a forward robustness in Appendix C.5.[1]

Table 4 reports the implied rate change for the merged Truist at the 2019 baseline and at each post-merger year. Column Δr_{merged} is the difference $r_{\text{merged}} - r_{\text{separate}}$ at that year's market state, with the un-merger counterfactual re-solved each year using observed Truist rates and shares to recover marginal costs. Negative values indicate the merger lowered Truist's deposit rate relative to the separate-ownership counterfactual (anti-competitive); positive values indicate the merger raised rates.

Time pattern of the implied rate effect. The headline 2019 column estimates the merger's implied effect using the actual pre-merger market state in the last year before consummation; this is our preferred number because the demand parameters are estimated on data that ends just before the panel year being simulated, so the fixed-$\hat{\alpha}$ assumption is least demanding. At this baseline the merger lowered Truist's deposit rate by approximately 25 basis points relative to the un-merger counterfactual, an anti-competitive effect consistent with the reduced-form sign in the difference-in-differences. The forward columns (2020–2023) ask what the merger's effect would be if the

[1]The simulation is implemented using the `mergersim` package of **?**. Implementation details, including the recovery of marginal costs from observed prices and the attribution of post-merger Truist branches to BB&T and SunTrust legacy entities, are described in Appendix C.3.

pre-merger demand parameters described post-merger depositor behavior. The implied effect is anti-competitive in 2020 (−8.7 bp) and reverses sign by 2021 (+4.8 to +9.5 bp). We read the time pattern as suggesting that the local market-power channel that drives the 2019 result attenuates over the post-merger years, consistent with rival entry, consolidation among smaller competitors, and the post-2020 deposit-pricing environment in which the federal funds rate rose sharply and depositor sensitivity to rates may have shifted. The forward sims should therefore be read as exploratory: $\hat{\alpha}$ held fixed at the pre-merger value is a strong assumption out of sample, particularly for 2022–2023 when policy rates jumped from near zero to over five percent.

Demand-side robustness. Appendix Table C.4 reports the same counterfactual under alternative pre-merger demand windows, varying the start year from 2013 to 2016 with the end year held at 2019. The estimated demand parameter $\hat{\alpha}$ rises as the window narrows toward the merger consummation, but the implied counterfactual rate change is small and negative across all windows, consistent with elastic depositor demand attenuating the price effect of the ownership change.

Welfare and surplus. We report deposit-weighted welfare envelopes that multiply the model-implied rate change by observed bank deposits in each overlap county-bank cell. For Truist, the envelope producer-surplus gain is $+\sum_{j \in \text{Truist}} D_j \cdot \Delta p_j$, where $\Delta p_j = p_j^{\text{merged}} - p_j^{\text{separate}}$ is the simulated change in the price variable (positive Δp corresponds to a rate markdown by Truist). The envelope consumer-surplus change is the symmetric quantity over all banks. Magnitudes are in millions of dollars per year, summed across overlapping counties.

Interpretation. The structural counterfactual implies that the merger lowered Truist's deposit rate in overlap counties by approximately 25 basis points relative to a continued-separate-ownership counterfactual, transferring approximately $306 million per year to Truist as producer surplus while costing depositors approximately $556 million per year. Net welfare loss is approximately $250 million per year. The structural rate effect is an order of magnitude larger than the difference-in-differences estimate of approximately 1–2 basis points (the headline annual specification) or 7–9 basis points at peak in the quarterly event study. The two are consistent: the difference-in-differences identifies the differential effect on overlap relative to non-overlap counties, netting out general industry trends, while the structural counterfactual recovers the Truist-specific markdown in overlap counties directly. We treat the structural number as the policy-relevant magnitude and the difference-in-differences as a conservative lower bound that complements it. We report alternative pre-merger windows and the BLP-full benchmark in Appendix C.5.

Table 5: Implied surplus changes under the no-merger counterfactual: Money Market $10,000, 2019 baseline, demand window 2013–2019

	Headline IV log rival county footprint	Robustness IV log rival total branches
$\hat{\alpha}$ (pp)	5.93	3.92
$t(\hat{\alpha})$	3.41	2.98
KP-F	22.9	27.1
Δr_{merged} (bp)	−25.5	−37.9
Δ Producer surplus, Truist ($M / yr)	+306	+455
Δ Consumer surplus, overlap ($M / yr)	−556	−826
Net welfare ($M / yr)	−250	−371

Notes: Implied changes in consumer and producer surplus from the merger relative to the un-merger counterfactual, evaluated at the 2019 baseline. Magnitudes are in millions of dollars per year, summed across overlapping counties. Demand is estimated on the pre-merger 2013–2019 sample. The headline IV uses log rival county footprint as the excluded instrument; the robustness column uses log rival total branches. Both are positive and significant with first-stage Kleibergen–Paap F above the Stock–Yogo 10% threshold. Welfare envelopes use the deposit-weighted formula $\sum_j D_j \cdot \Delta p_j$ with Δp in decimal-fraction units, which is the unit-invariant mechanical consequence of the simulated rate change. Implementation details are in Appendix C.3.

6 Conclusion

This paper studies the competitive effects of the 2019 BB&T–SunTrust merger in U.S. retail banking. The partial geographic overlap between the two banks generates clean variation that separates the local competition effect of increased concentration from the scale effect of becoming a larger institution. We bring together a reduced-form difference-in-differences design and a structural deposit-demand-and-pricing model to quantify the merger's pricing and stability consequences.

The reduced-form analysis shows that, in counties where both BB&T and SunTrust operated before the merger, rival banks widened their deposit spreads by approximately two basis points on the headline annual specification and by roughly seven to nine basis points at peak in the quarterly event study, both relative to rivals in counties served by only one of the two predecessors. The effect is statistically significant in the unweighted county-and-year fixed-effect specification and persists through the end of our sample period. The branch-level robustness using only RateWatch data shows that the merger's effect is particularly large in counties without competing top-four U.S. banks, where deposit spreads rose by approximately fifteen basis points. These magnitudes are at the lower end of the retail-banking merger-retrospective literature and consistent with both bank-wide pricing rules at large institutions and elastic depositor demand for liquid deposit products.

The structural counterfactual reassigns Truist's branches to their 2019 BB&T or SunTrust legacy ownership, holding pre-merger demand parameters fixed at the values estimated on the 2013–2019 sample under the single-IV specification (log rival county footprint as the excluded instrument). The implied rate effect is anti-competitive in sign and economically meaningful: at

the 2019 baseline the merger lowered Truist's deposit rate by approximately 25 basis points relative to the un-merger counterfactual. The implied change in consumer surplus across overlap counties is a loss of approximately $556 million per year, and the implied producer-surplus gain to Truist is approximately $306 million per year. The consumer loss exceeds the producer gain because rival banks in overlap counties also lower rates in equilibrium, so depositors at non-Truist banks are also affected. The structural rate effect is an order of magnitude larger than the difference-in-differences estimate; the gap is consistent with the DiD netting out general industry trends that the structural model captures directly.

The bank-level stability event study shows that the merged Truist experienced a short-run drop in its log Z-score around the merger consummation, contemporaneous with the COVID-19 pandemic, followed by a recovery to its pre-merger trend by mid-2021. We treat this evidence as a useful descriptive complement rather than a clean causal estimate, and we report robustness across alternative treatment definitions in Appendix B.5.

The results indicate that a major regional bank merger of this scale produced modest but detectable anti-competitive effects on deposit pricing and a temporary, mostly mechanical disruption to balance-sheet stability that recovered within two years. From a policy standpoint, the heterogeneity of effects across competitive environments, with larger spreads in markets without top-four bank competition, suggests that antitrust review of bank mergers should weight overlapping counties' outside competitive options more heavily than aggregate concentration measures alone.

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Appendix

A Additional Descriptive Statistics

This appendix provides supplementary descriptive evidence on the U.S. banking industry that complements the regression-sample summary in the main text.

A.1 Universe of U.S. commercial banks (full SoD snapshot)

For reference, Table A.1 reports the full universe of U.S. commercial banks from the FDIC Summary of Deposits, broken down by AUM tier (Panel A), state footprint (Panel B), and county footprint among single-state banks (Panel C). The regression-sample restriction to banks observed in the RateWatch panel for the Money Market $10,000 product (Table 1) drops the firm count from 4,649 to 1,854 but preserves the broad tier structure.

Table A.1: Background statistics for U.S. banks (full SoD universe, June 2023)

Panel A. Bank size by assets under management (AUM)				
Bank AUM	Number of Firms	Number of Branches	Deposits Held (billions)	Deposit Share
$1+ T	4	13,911	$6,050	35.1%
$250 B – $1 T	10	9,303	$3,236	18.8%
$100 B – $250 B	19	8,049	$2,446	14.2%
$10 B – $100 B	125	13,060	$2,656	15.4%
$1 B – $10 B	842	18,014	$1,910	11.1%
≤ $1 B	3,655	15,459	$953	5.5%
Total	**4,649**	**77,474**	**17,169**	**100%**
Panel B. Bank size by state footprint				
Branch Footprint	Number of Firms	Number of Branches	Deposits Held (billions)	Deposit Share
25+ states	5	18,024	$6,269	36.5%
10-24 states	23	14,498	$3,103	18.1%
2-9 states	703	22,713	$4,127	24.0%
1 state	3,918	22,239	$3,671	21.4%
Panel C. County footprint of banks operating in one state				
Branch Footprint	Number of Firms	Number of Branches	Deposits Held (billions)	Deposit Share
10+ counties	88	3,357	$339	9.2%
2-9 counties	2,140	15,284	$1,442	39.3%
1 county	1,690	3,598	$1,890	51.5%

Notes: Data as of June 30, 2023. Deposit share is computed as a fraction of total U.S. commercial bank deposits in Panels A and B and relative to single-state banks only in Panel C. Branch footprint is defined by the number of states in which a bank operates at least one branch in Panel B and the number of counties in which a bank operates at least one branch in Panel C. Panel C is restricted to banks operating in a single state. Data source: Summary of Deposits (FDIC).

B Reduced-form Robustness and Auxiliary Specifications

This appendix collects robustness exercises for the reduced-form deposit-spread results: alternative treatment contrasts, the bank-county fixed-effects specification, the CD product, the branch-level RateWatch panel, and alternative treatment definitions for the bank-level stability event study.

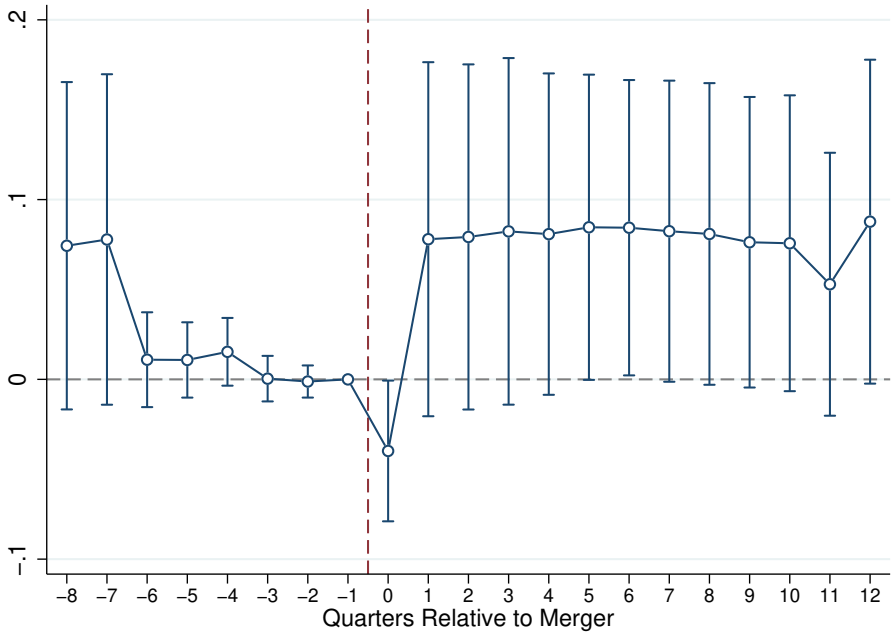


Figure B.1: Quarterly event study: deposit spread, Both vs. One, Money Market $10,000

Notes: Quarterly event-study coefficients from equation (3) on the bank–county–quarter market panel for the Money Market $10,000 product. Sample is rivals of BB&T and SunTrust in counties classified as Both vs. One on the basis of pre-merger Truist-pair presence. Treatment is the interaction between the bank-county's pre-merger Truist-pair classification and event-time relative to merger consummation in 2019Q4. The omitted reference is event-time −1 (2019Q3); the implementation uses the placement of the omitted dummy at the end of the regression to let the collinearity-drop algorithm select the reference, producing a zero anchor at quarter −1. The vertical dashed line marks merger consummation. The specification absorbs bank–county and quarter fixed effects with the bank and market controls used in the annual specification. Standard errors are clustered at the bank–county level. Ninety-five-percent confidence intervals are shown. Data sources: RateWatch, Summary of Deposits, Federal Reserve Economic Data, FFIEC Call Reports.

B.1 Reduced-form deposit-spread results, alternative contrasts and products

Table B.5 reports the corresponding branch-level difference-in-differences estimates for the auxiliary contrasts. The branch-level specification uses the raw RateWatch panel without aggregation to the bank-county-year cell and includes only month and county fixed effects (no bank or market controls), so the table exposes the competitive contrast more directly than the controlled market-panel regressions in the main text. The three auxiliary contrasts isolate different margins of the merger: One vs. None compares counties served by exactly one of the two predecessors to counties served by neither, isolating the bank-wide scale effect of merging into a larger institution; Any vs. None compares counties served by either predecessor to counties served by neither, capturing the joint effect of geographic footprint; Both vs. None compares overlapping counties to counties served by neither predecessor, combining local competition and scale margins. Across these contrasts the point estimates have the same sign as the headline Both vs. One specification but consistently smaller magnitudes and standard errors that overlap zero, confirming that the within-Truist contrast in the main text is the more informative comparison.

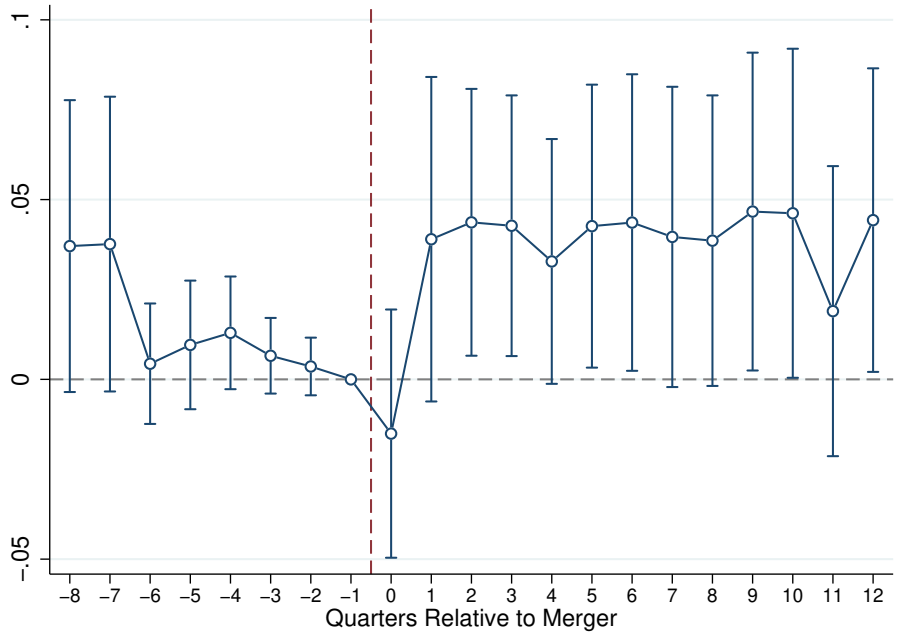


Figure B.2: Quarterly event study: deposit spread, Both vs. One, Money Market $2,500

Notes: Same specification as Figure B.1, applied to the Money Market $2,500 product. Sample, fixed effects, controls, treatment definition, and clustering structure are unchanged. The smaller post-merger magnitudes relative to the $10,000 product are consistent with greater depositor rate sensitivity at the lower-balance tier. Data sources: RateWatch, Summary of Deposits, Federal Reserve Economic Data, FFIEC Call Reports.

B.2 Annual market-panel DiD with bank-county fixed effects

Table B.1 repeats the headline specification with bank-county and year fixed effects in place of the headline's county and year fixed effects, isolating only within-bank-county time variation. Two features of the deposit market compress this within-bank-county variation. First, both BB&T/SunTrust and most large rivals operating in the same counties (Bank of America, JPMorgan Chase, Wells Fargo, Citi) post a single rate across many counties under zone-based pricing, so the same bank's spread in an overlapping county and a single-Truist county tends to move together over time. Second, the within-bank-county time series is short: the 2016–2022 window provides only seven years, of which 2020 is dropped as a transition year, leaving a two-year post-merger window (2021 and 2022). Table B.2 extends the comparison to the twelve-month $2,500 CD product, where the merger effect is large enough to remain significant under bank-county fixed effects.

B.3 Robustness across products: CD 12-month $2,500

Table B.2 repeats the two-column specification on the twelve-month CD $2,500 product. Columns (1) and (2) report the headline county and year fixed-effects specification; columns (3) and (4) report the bank-county and year fixed-effects specification. The point estimates are larger than for the headline Money Market $10,000 product (about seven basis points under county and year fixed effects, and four basis points under the strict bank-county fixed effects), and the merger effect remains statistically significant in both specifications. The result confirms that the merger's

Table B.1: Both-vs-One Annual Market-Panel DiD, Money Market $10,000 (bank-county and year fixed effects)

	(1) FE only	(2) + controls
Treat × Post	0.008	0.012
	(0.007)	(0.007)
Bank-county FE	✓	✓
Year FE	✓	✓
Bank + Market Controls		✓
Observations	6,786	6,786

Notes: Annual market-panel DiD for the Both vs. One contrast on the Money Market $10,000 product, with bank-county and year fixed effects in place of the headline's county and year fixed effects. Sample, dependent variable, treatment definition, controls, and clustering match Table 3; the smaller observation count (6,786 vs. 6,817) reflects 31 singleton bank-county cells dropped by `reghdfe` when the bank-county fixed effect is added. The bank-county fixed effect isolates within-bank-county time variation. Standard errors clustered at the bank-county level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Both-vs-One Annual Market-Panel DiD, CD 12-month $2,500 (county-year vs. bank-county fixed effects)

	County and year FE		Bank-county and year FE	
	(1) FE only	(2) + controls	(3) FE only	(4) + controls
Treat × Post	0.020	0.068***	0.024	0.041**
	(0.024)	(0.023)	(0.020)	(0.018)
County FE	✓	✓		
Bank-county FE			✓	✓
Year FE	✓	✓	✓	✓
Bank + Market Controls		✓		✓
Observations	5,593	5,593	5,558	5,558

Notes: Annual market-panel difference-in-differences estimates for the Both vs. One contrast on the twelve-month $2,500 CD product. Sample, dependent variable, and treatment definition match Table 3: bank–county–year cells for 2016–2022, restricted to rivals of BB&T and SunTrust; deposit spread (federal funds rate minus deposit rate, in percentage points) as the outcome; Treat equal to one in overlapping counties; Post equal to one for years 2021 onward, with 2020 dropped. Columns (1) and (2) use the headline county-and-year fixed-effects specification (no bank fixed effects), so the treatment effect is identified from both cross-bank variation within counties and within-bank-county time variation. Columns (3) and (4) replace the county fixed effect with a bank-county pair fixed effect, isolating only within-bank-county time variation. Bank and market controls (added in columns 2 and 4) match those in Table 3. Standard errors are clustered at the bank-county level. Sample size differs slightly between the two FE structures because singleton bank-county pairs are dropped in the bank-county fixed-effect estimation. Data sources: RateWatch, Summary of Deposits, Federal Reserve Economic Data, FFIEC Call Reports. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

effect on deposit spreads is robust to the choice of product and to the strict within-bank-county identification.

Table B.3: Difference-in-differences results, branch level (RateWatch data only)

	(1)	(2)	(3)	(4)
Treat × Post	0.044***	0.004	0.013	0.152***
	(0.014)	(0.008)	(0.009)	(0.026)
Month FE		✓	✓	✓
County FE		✓	✓	✓
Top-4 competitor sample			✓	
No top-4 competitor sample				✓
Observations	171,284	171,277	149,738	21,539

Notes: Branch-month-level DiD using only RateWatch data, no bank or market controls. Column (1) is OLS without fixed effects. Column (2) adds month and county fixed effects. Column (3) restricts the sample to counties with at least one top-4 U.S. bank (JPMorgan Chase, Bank of America, Citi, Wells Fargo) operating in the county. Column (4) restricts to counties without any top-4 U.S. bank. Standard errors are clustered at the county level. The contrast between columns (3) and (4) is the heterogeneity-by-outside-competition result discussed in Section 5.3: the merger raised spreads substantially in markets without competition from a top-4 bank, while leaving spreads largely unchanged in markets with strong outside competition.
Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.4 Branch-level robustness using RateWatch only

As a robustness check we estimate the difference-in-differences specification at the branch–month level using the raw RateWatch panel without aggregating to the bank-county-year market panel and without the bank or market controls used in the main specification. The dependent variable is the deposit spread on the Money Market $10k product. Treat indicates counties with both BB&T and SunTrust prior to the merger, and Post equals one for periods after December 2019. Standard errors are clustered at the county level.

B.5 Stability event study, alternative treatment definitions

Figure B.3 shows the headline log Z-score event study using the combined Truist-pair indicator (BB&T pre, SunTrust pre, Truist post). Figure B.4 reports the same event study with two alternative treatment definitions: BB&T-only (the legal entity that becomes Truist post-merger) and SunTrust-only (the predecessor that ceases to exist after consummation, so post-event coefficients are mechanically undefined). The post-merger pattern is similar across the first two definitions, supporting the interpretation in the main text that the short-run drop and medium-run recovery are properties of the merged institution rather than of one predecessor alone.

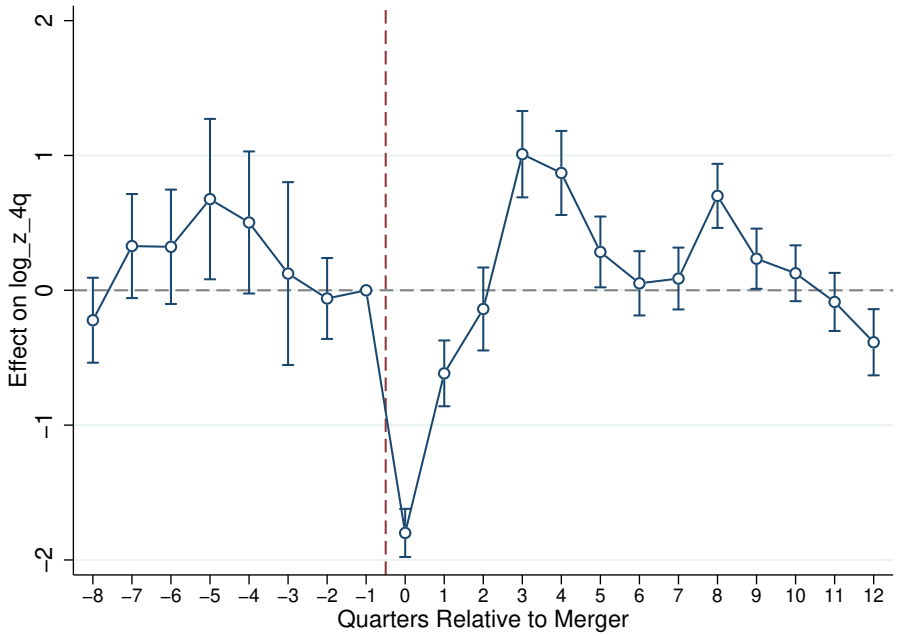


Figure B.3: Bank-level stability event study: $\log(Z_{4q})$, combined Truist-pair treatment

Notes: Event-study coefficients from equation (5). Treatment is BB&T pre-merger, SunTrust pre-merger, or Truist Bank post-merger. Control group is regional banks with 2019 assets in \$10B–\$1T (n=128). Bank and quarter fixed effects, controlling for the equity-to-assets ratio. Standard errors clustered at the bank level. The vertical dashed line marks merger consummation in 2019Q4. Data source: Call Reports.

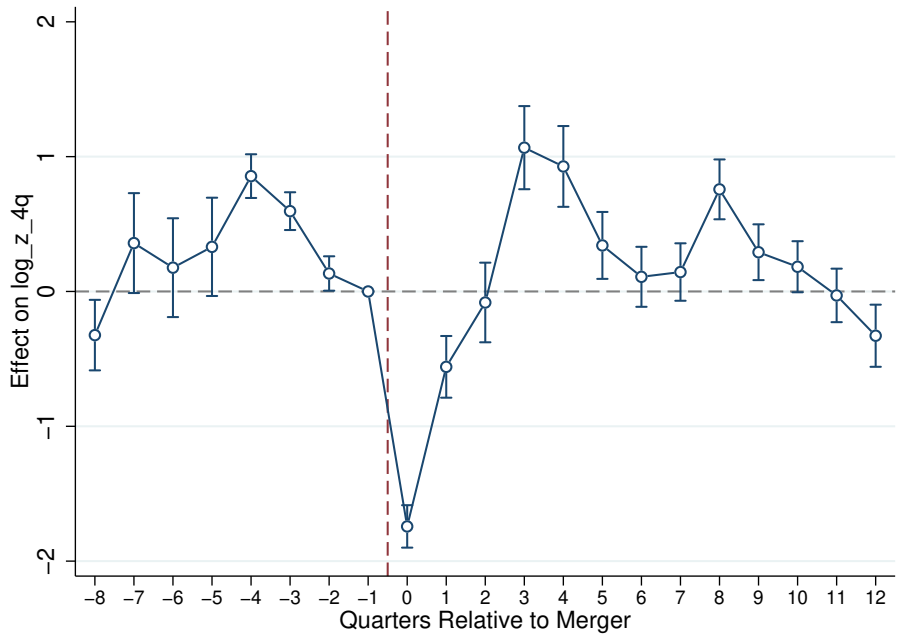


Figure B.4: Bank-level stability event study: BB&T continuing legal entity

Notes: Event-study coefficients from equation (5) with the treatment indicator restricted to the BB&T charter, which is the surviving legal entity that continues as Truist Bank after the merger. Specification matches Figure B.3 (bank and quarter fixed effects, equity-to-assets-ratio control, standard errors clustered at the bank). The control group is the regional-bank peer panel of banks with 2019 assets in the \$10 billion to \$1 trillion range. The omitted period is event-time −1; the vertical dashed line marks merger consummation in 2019Q4. We do not report a SunTrust-only counterpart because the SunTrust legal entity was dissolved at consummation, so post-merger coefficients for that treatment are mechanically zero. Data source: FFIEC Call Reports.

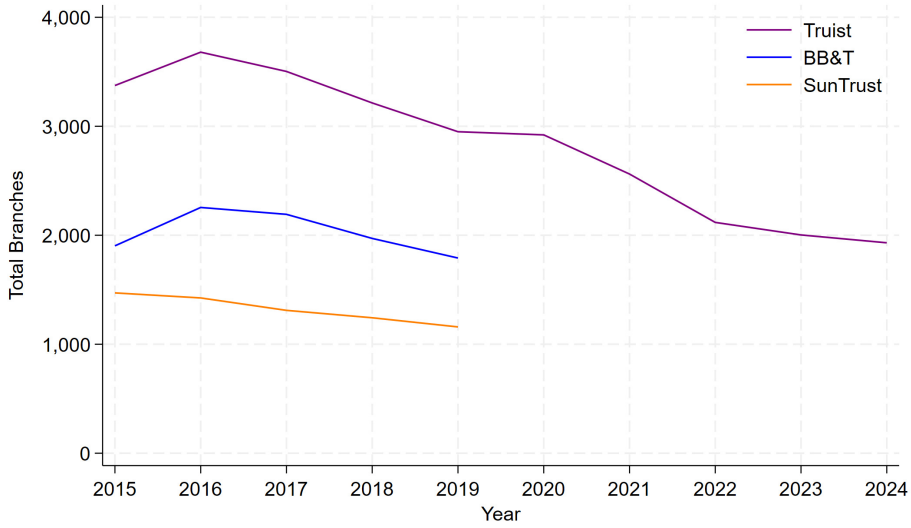


Figure B.5: Total branches at SunTrust, BB&T, and Truist

Notes: Total branch counts by year for SunTrust, BB&T, and the combined Truist. The merger closed in December 2019. Subsequent branch consolidation is reflected in the declining count for Truist after 2020. Data source: Summary of Deposits.

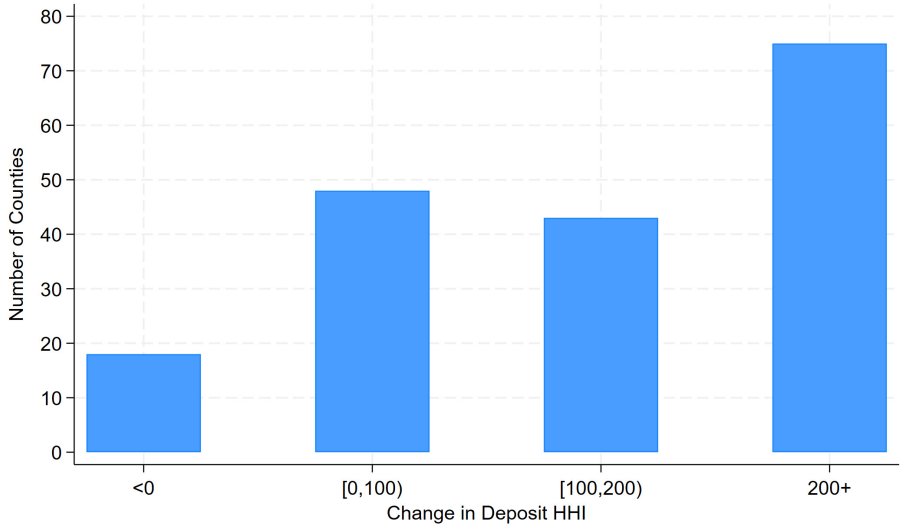


Figure B.6: Change in deposit HHI in overlapping counties

Notes: Distribution of the change in the Herfindahl-Hirschman Index (HHI) across counties that had branches of both BB&T and SunTrust before the merger. HHI is computed using branch-level deposit shares from the Summary of Deposits (FDIC). This change is calculated from comparing HHI in June 30, 2020, (the year after the merger) against June 30, 2019 (just before the merger). A larger increase indicates a greater reduction in local competition.

Table B.4: Summary statistics and balance tests for treatment and control groups

	Panel A. Summary Statistics			
	Both	One	Any	None
Deposit Spread	0.929	0.931	0.930	0.957
County HHI	852	1095	987	1421
County Branches	190	202	197	130
County Firms	30.5	33.5	32.2	21.2
Observations	75,053	93,352	168,405	648,865
	Panel B. Differences in Means			
	Both vs. One	One vs. None	Any vs. None	Both vs. None
Deposit Spread	0.002	0.026***	0.027***	0.028***
	(0.007)	(0.005)	(0.004)	(0.005)
County HHI	243***	326***	435***	569***
	(5.572)	(4.877)	(3.769)	(5.440)
County Branches	12.6***	-72.5***	-66.9***	-59.9***
	(1.203)	(1.071)	(0.813)	(1.154)
County Firms	3.0***	-12.3***	-11.0***	-9.3***
	(0.137)	(0.092)	(0.070)	(0.094)
Observations	168,405	742,217	817,270	723,918

Notes: This table reports summary statistics and pre-merger balance across treatment and control groups for each specification. Deposit spread is defined as the federal funds rate minus the deposit rate. County HHI is computed using branch-level deposit shares from the Summary of Deposits (FDIC). County branches and firms measure the number of bank branches and distinct banking firms operating in each county. Treatment groups are defined based on pre-merger presence of BB&T and SunTrust as described in Table 2. Data sources: RateWatch, Summary of Deposits.

Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Branch-level difference-in-differences: auxiliary contrasts (Money Market $10,000)

	(1)	(2)	(3)	(4)
	Panel A. One vs. None			
Treat × Post	-0.065***	-0.008	-0.027***	0.078***
	(0.010)	(0.005)	(0.006)	(0.010)
Observations	761,250	761,227	445,534	315,693
	Panel B. Any vs. None			
Treat × Post	-0.046***	-0.006	-0.021***	0.103***
	(0.008)	(0.004)	(0.005)	(0.009)
Observations	837,575	837,549	518,448	319,101
	Panel C. Both vs. None			
Treat × Post	-0.021**	-0.003	-0.014**	0.227***
	(0.011)	(0.006)	(0.007)	(0.023)
Observations	742,616	742,594	441,624	300,970
Month FE		✓	✓	✓
County FE		✓	✓	✓
Top-4 competitor sample			✓	
No top-4 competitor sample				✓

Notes: Branch-level difference-in-differences estimates on the raw RateWatch panel (no aggregation to the bank-county-year cell) for the Money Market $10,000 product. Sample period 2016–2022, with 2020 dropped as a transition year. Three panels report alternative treatment-control comparisons: One vs. None compares counties served by exactly one of BB&T or SunTrust before the merger to counties served by neither, isolating the bank-wide scale effect; Any vs. None compares counties served by either predecessor to counties served by neither, capturing geographic-footprint exposure; Both vs. None compares overlapping counties to counties served by neither predecessor, combining local-competition and scale margins. Dependent variable is the deposit spread (federal funds rate minus the deposit rate, in percentage points). Column (1) is the OLS specification with no fixed effects; column (2) adds month and county fixed effects; columns (3) and (4) split the sample by the presence of a top-four U.S. bank (JPMorgan Chase, Bank of America, Citi, Wells Fargo) operating in the county. Standard errors are clustered at the county level and reported in parentheses. The leaner control set (only month and county fixed effects) and the larger branch-level sample size differ from the headline market-panel specification in Table 3. Data sources: RateWatch, Summary of Deposits.
Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: Pre-merger Comparison: Truist vs. Control Banks

	Truist	Control	Difference	t-stat
Z-score	246.904	402.008	-155.105***	-3.30
Log(Z-score)	5.194	5.896	-0.702***	-4.19
Return on assets	0.003	0.003	0.000	0.16
Equity-to-assets ratio	0.124	0.125	-0.001	-0.13
SD(ROA), 8q	0.001	0.000	0.001***	3.46
Assets (billions)	141	206	-65***	-11.04
Observations	38	314		
Banking firms	2	21		

Notes: This table reports pre-merger summary statistics for Truist banks (SunTrust, BB&T, and Truist Bank) and a tighter pre-merger level-balance control group of regional banks with 2019 assets between $100 billion and $250 billion (n=21 banks), distinct from the broader $10 billion to $1 trillion event-study control panel of 128 banks used in Section 3.5. The sample is restricted to observations prior to the Truist merger (2019Q4). The difference column reports the difference in means (Truist minus Control), and the t-statistic is from a two-sample t-test. Z-scores in this balance table are constructed using an eight-quarter rolling window for the standard deviation of return on assets to smooth the pre-merger level comparison; the event-study analysis in Section 3.5 uses a four-quarter window. Data source: Call Reports.
Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.1: Demand estimates: Money Market $10,000, 2013–2019

	(1) IV (headline)	(2) IV + Cost	(3) Nested-logit IV
Money-market rate	5.927***	8.620***	8.452***
	(1.736)	(0.643)	(0.615)
Log within-nest share			0.296***
			(0.024)
Log branches in county	0.726***	0.789***	0.703***
	(0.050)	(0.038)	(0.038)
Log counties served	0.131***	0.185***	0.162***
	(0.035)	(0.014)	(0.014)
County branch share	4.252***	4.111***	2.946***
	(0.181)	(0.190)	(0.195)
Large bank indicator	0.108***	0.094***	0.183***
	(0.019)	(0.023)	(0.024)
Log Z-score (4-quarter avg.)	0.010*	0.017**	0.011*
	(0.006)	(0.007)	(0.006)
Fixed effects	county_year	county_year	county_year
Excluded IV set	ln rival county footprint	+ cost shifters	+ cost shifters
Kleibergen-Paap Wald F	22.86	34.10	33.84
Observations	59,360	59,304	59,304

Notes: Logit IV demand estimates for the Money Market $10,000 product, estimated on the bank-county-year market panel for 2013–2019 (pre-merger). Dependent variable: logit mean utility, $\delta_{jmt} = \log s_{jmt} - \log s_{0mt}$. Bank and market controls are the log of bank-j branches in county m, the log of the number of counties in which bank j operates, the within-county branch share, an indicator for banks with assets above $10 billion, and the log Z-score (4-quarter rolling) as a stability control. Column (1) is the headline IV specification using a single just-identified instrument: the leave-one-out log of the number of counties served by rival banks in the same market ("log rival county footprint"). Columns (2) and (3) add the equity-to-assets ratio, log non-interest expense over assets, and log assets per employee as cost-shifter instruments; column (3) further adds the log within-nest share as a second endogenous variable in a nested-logit specification. All specifications include county and year fixed effects. Standard errors clustered at the county-year level. We do not report an OLS benchmark or the Hansen J over-identification statistic: OLS is severely downward-biased in this setting because banks adjust rates in response to local demand shocks; the Hansen J rejects in the over-identified specifications, consistent with mild misspecification of the cost-shifter channel. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C Structural Model: Implementation and Robustness

This appendix reports the full demand-estimation tables, implementation details for the no-merger counterfactual, robustness across alternative pre-merger demand windows, and a forward counterfactual for the savings product.

C.1 Demand estimation table

Table C.1 reports the headline demand specification for the Money Market $10,000 product referenced throughout the structural section. The table reports three columns: an instrumental-variables specification using BLP-style rival sums as excluded instruments, the IV specification augmented with cost shifters, and a nested-logit version with county-defined nests.

C.2 Demand estimation robustness across windows and specifications

Table C.2 reports the demand estimates for the headline Money Market $10,000 product across alternative pre-merger windows (varying the start year) and across the three IV specifications (single-IV headline, IV plus cost shifters, nested-logit IV with within-nest log share). The own-rate coefficient $\hat{\alpha}$ is positive and significant in every cell, with magnitudes in the 4–9 pp range that are stable to the choice of window and specification. The Kleibergen–Paap first-stage F exceeds the Stock–Yogo 10% threshold (approximately 16) in every cell.

Table C.2: Demand estimates: robustness across pre-merger windows and IV specifications, Money Market $10,000

	2013–2019	2014–2019	2015–2019	2016–2019
Panel A: Headline IV (single rival county footprint)				
Rate coef.	5.927***	6.006***	4.776***	3.713***
(SE)	(1.736)	(1.881)	(1.583)	(1.269)
Log branches in county	0.726***	0.721***	0.698***	0.685***
KP–F	22.86	19.19	18.27	21.22
Obs	59,360	50,229	41,288	32,696
Panel B: IV + Cost shifters				
Rate coef.	8.620***	7.130***	8.496***	8.457***
(SE)	(0.643)	(0.475)	(0.639)	(0.755)
Log branches in county	0.789***	0.746***	0.793***	0.831***
KP–F	34.10	34.77	23.33	17.70
Obs	59,304	50,180	41,244	32,661
Panel C: Nested-logit IV				
Rate coef.	8.452***	6.621***	7.940***	7.961***
(SE)	(0.615)	(0.428)	(0.580)	(0.689)
Log branches in county	0.703***	0.657***	0.699***	0.733***
KP–F	33.84	34.77	23.17	17.33
Obs	59,304	50,180	41,244	32,661

Notes: Logit IV demand estimates for the Money Market $10,000 product, varying the pre-merger demand window (start year) across columns and the IV specification across panels. Panel A is the headline single just-identified instrument (leave-one-out log of the number of counties served by rival banks in the same market). Panel B adds the equity-to-assets ratio, log non-interest expense over assets, and log assets per employee as cost-shifter instruments. Panel C adds the log within-nest share as a second endogenous variable in a nested-logit specification, with nests defined by the large-bank indicator. All specifications include county and year fixed effects and the same bank-network controls as Table C.1. Standard errors clustered at the county-year level. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Counterfactual implementation

The counterfactual simulation is implemented with the `mergersim` package of **?**, which inverts the multinomial-logit first-order conditions to recover firm-level marginal costs from observed prices and shares and then solves the post-merger Bertrand–Nash equilibrium under the counterfactual ownership structure. We hold the demand parameters fixed at the pre-merger estimates and recover marginal costs from observed Truist prices using the merged-firm first-order conditions at each post-merger year. The no-merger equilibrium is solved by re-imposing separate BB&T and SunTrust ownership through the `newfirm()` option of `mergersim simulate`. Each post-merger Truist branch is attributed to its 2019 legacy entity (BB&T or SunTrust) using `Summary of Deposits` branch

records: a branch is assigned to the predecessor that owned it in the June 2019 SOD release. County-year cells in which both predecessors had branches are split proportional to their 2019 deposit shares.

C.4 Forward counterfactual by product

Table C.3 reports the no-merger counterfactual for the savings $2,500 product.

Table C.3: No-merger counterfactual by year, Savings $2,500

	2020	2021	2022	2023
Δ rate, merged (pp)	0.0073	0.0072	0.0063	0.0031
Δ rate, rivals (pp)	0.0026	0.0025	0.0024	0.0009
N obs	34,155	34,155	34,155	34,155
Converged	1	1	1	1

Notes: No-merger counterfactual for the Savings $2,500 product, with one column per post-merger year. Demand parameters are estimated on the pre-merger 2013–2019 sample using the same single-IV specification as the headline Money Market $10,000 demand model and held fixed across the post-merger years. Marginal costs are recovered from observed Truist post-merger prices and shares using the merged-firm Bertrand–Nash first-order conditions at year t, and the counterfactual equilibrium is solved under separate BB&T and SunTrust ownership using the legacy attribution from 2019 `Summary of Deposits` branch records. Reported quantities: implied counterfactual rate change for the merged firm and for non-merging rivals (in percentage points), the simulation sample size, and a convergence indicator. Implementation details are in Appendix C.3. Implied rate effects on the savings product are small and slightly positive, indicating a near-zero competition effect of the merger in this product. Data sources: RateWatch, Summary of Deposits, FFIEC Call Reports.

C.5 Counterfactual robustness across pre-merger demand windows

Table C.4 reports the no-merger counterfactual for the Money Market $10,000 product when the pre-merger demand sample is varied across alternative start and end years.

Table C.4: Merger simulation results by pre-merger window, Money Market $10,000

	2013-2019	2014-2019	2015-2019	2016-2019	2016-2020	2016-2021	2016-2022	2016-2023
Δ rate, merged (pp)	-0.0088	-0.0093	-0.0122	-0.0252	-0.0232	-0.0103	-0.0113	-0.0070
Δ rate, rivals (pp)	-0.0002	-0.0002	-0.0003	-0.0007	-0.0006	-0.0003	-0.0003	-0.0002
N obs	63,163	53,428	43,982	34,850	42,974	50,789	58,417	65,912
Converged	1	1	1	1	1	1	1	1

Notes: No-merger counterfactual for the Money Market $10,000 product. Each column corresponds to a different pre-merger demand-estimation sample. Start years vary from 2013 to 2016 with the end year held at 2019, and end years vary from 2019 to 2023 with the start year held at 2016. Demand parameters are estimated on the indicated window using the headline single-IV specification of Table C.1 and held fixed when computing the counterfactual at the 2019 baseline. Marginal costs are recovered from observed Truist post-merger prices and shares using the merged-firm Bertrand–Nash first-order conditions; the counterfactual is solved with separate BB&T and SunTrust ownership using the legacy attribution from 2019 `Summary of Deposits` branch records. Reported quantities: implied rate change for the merged firm and for rivals (in percentage points), the simulation sample size, and a convergence indicator. Implementation details are in Appendix C.3. Data sources: RateWatch, Summary of Deposits, FFIEC Call Reports.